



Mini Project Report On

IoT Framework for Helmet Theft Prevention Using Sensor Fusion and Behavioral Anomaly Detection

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By

Gautham Girish (RET23CS102)

Gautham Jayesh (RET23CS103)

Kurian Kochikunnel (RET23CS136)

Maneesh James Antony (RET23CS144)

Under the guidance of

Mr. Harikrishnan M

**Department of Computer Science & Engineering
Rajagiri School of Engineering & Technology (Autonomous)
(Affiliated to APJ Abdul Kalam Technological University)**

Rajagiri Valley, Kakkanad, Kochi, 682039

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CERTIFICATE

*This is to certify that the mini project report entitled "**IoT Framework for Helmet Theft Prevention Using Sensor Fusion and Behavioral Anomaly Detection**" is a bonafide record of the work done by **Gautham Girish (RET23CS102)**, **Gautham Jayesh(RET23CS103)**, **Kurian KochiKunnel (RET23CS136)**, **Maneesh James Antony (RET23CS144)**, submitted to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2025-2026.*

Project Guide

Mr Harikrishnan M

Assistant Professor

Dept. of CSE

RSET

Project Coordinator

Mrs Seema Safar

Assistant Professor

Dept. of CSE

RSET

Dr Preetha K G
Head of the Department
Dept. of CSE
RSET

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Gautham Girish

Gautham Jayesh

Kurian Kochikunnel

Maneesh James Antony

Abstract

Helmet Theft is a common issue in public parking areas, leading to financial loss and inconvenience for users. This project presents the development of a Smart System designed to detect the unauthorized movement of a helmet. The system integrates an ESP32 microcontroller, a motion sensor MPU6050, and a GSM communication model to provide an automated security solution.

The MPU6050 sensor continuously monitors the motion and orientation of the helmet. When abnormal movement indicating possible theft is detected, the ESP32 processes the sensor data and triggers the alert mechanism. The GSM module sends an alert message informing the user about the theft, allowing them to take appropriate steps.

Compared to traditional helmet security methods, such as manual locks, the proposed system provides automatic detection and remote alert notification, making it a more effective and intelligent solution. The compact design and the use of low-power components allow the system to be embedded inside a helmet without affecting usability.

Overall, the project demonstrates a practical application of IoT technologies for enhancing personal security and to prevent theft.

Contents

Acknowledgements	i
Abstract	ii
List of Figures	v
List of Tables	vi
List of Abbreviations	vii
1 Introduction	1
1.1 Background	1
1.2 Problem Definition	1
1.3 Scope and Motivation	2
1.3.1 Scope	2
1.3.2 Motivation	2
1.4 Objectives	2
1.5 Challenges	3
1.6 Assumptions	3
1.7 Societal / Industrial Relevance	3
1.8 Organization of the Report	3
2 Software Requirements Specification	5
2.1 Introduction	5
2.2 Overall Description	5
2.3 External Interface Requirements	6
2.4 System Features	6
2.5 Other Nonfunctional Requirements	7

3	System Architecture and Design	8
3.0.1	System Overview	8
3.0.2	Architectural Design	8
3.0.3	Proposed Methodology/Algorithms	9
3.1	User Interface Design	10
3.2	Database Design	11
3.3	Description of Implementation Strategies	12
3.4	Module Division	13
4	Results and Discussions	15
4.1	Overview	15
4.2	Testing	15
4.3	Quantitative Results	16
4.4	Discussion	17
5	Conclusion	18
5.1	Conclusion	18
5.2	Future Scope	18
	References	20
	Appendix A: Presentation	22
	Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes	31
	Appendix C: CO-PO-PSO Mapping	36

List of Figures

3.1	System Architecture Diagram	8
3.2	Circuit Diagram	9
5.1	Introduction Slide	23
5.2	Problem Definition Slide	23
5.3	Objectives Slide	24
5.4	Scope and Relevance Slide 1	24
5.5	Scope and Relevance Slide 2	25
5.6	System Design Slide 1	25
5.7	System Design Slide 2	26
5.8	System Architecture Slide	26
5.9	Hardware System Architecture Slide	27
5.10	Work Division Slide	27
5.11	Software Requirements Slide	28
5.12	Hardware Requirements Slide	28
5.13	References Slide	29
5.14	Results Slide	29
5.15	Conclusion Slide	30
5.16	Future Enhancements Slide	30

List of Tables

2.1	Hardware Interfaces	6
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List of Abbreviations

IoT - Internet of Things

GSM - Global System for Mobile Communication

MPU6050 - Motion Processing Unit 6050

SMS - Short Message Service

SRS - Software Requirements Specification

IDE - Integrated Development Environment

UART - Universal Asynchronous Receiver-Transmitter

I2C - Inter-Integrated Circuit

SIM - Subscriber Identity Module

GPRS - General Packet Radio Service

GPS - Global Positioning System

Chapter 1

Introduction

1.1 Background

In recent years, the number of two-wheelers used for daily transportation has increased significantly. As a result, helmets have become an essential safety accessory for riders and are mandatory in many regions. Riders often leave their helmets on their motorcycles due to the inconvenience of carrying them, making them easy targets for theft

Traditional security methods such as manual helmet locks and surveillance cameras offer limited protection and do not provide a reliable way to recover a stolen helmet. With the rapid development of embedded systems and IoT technologies, it is now possible to design smart security solutions that monitor and protect personal belongings. IoT systems allow sensors, microcontrollers, and communication modules to work together to detect events, process data, and send real time alerts to users.

This project proposes the development of a Smart Helmet Anti-Theft System that can detect unauthorized movement of a helmet and notify the owner. The system integrates motion sensing, and GSM communication to provide an effective and automated security solution. By combining these technologies, the system aims to improve helmet security and increase the chances of recovering stolen helmets

1.2 Problem Definition

Helmet theft is a common issue due to the absence of effective monitoring systems when helmets are left unattended. The aim of this project is to design a smart system that can detect unauthorized helmet movement and alert notifications to the owner

1.3 Scope and Motivation

1.3.1 Scope

The scope of this project is to design and implement a compact embedded system that can be integrated inside a helmet to detect theft. The system uses an ESP32 microcontroller along with sensors and communication modules to monitor helmet movement and transmit alerts. It focuses on detecting abnormal motion using a motion sensor and sending alerts through a GSM communication module to notify the user. The project focuses mainly on improving helmet security using low-cost and easily available components.

1.3.2 Motivation

Helmet theft is a frequent problem faced by many motorcycle users, especially in crowded public places. Losing a helmet not only causes financial loss but also creates inconvenience for riders. Existing security methods are either inconvenient or ineffective in preventing theft. The motivation behind this project is to create a simple and intelligent solution that can automatically detect theft and help locate the stolen helmet. By integrating IoT technologies with embedded systems, this project aims to enhance personal security and promote the use of smart devices for everyday problems

1.4 Objectives

1. To design a smart helmet anti-theft system using embedded hardware components.
2. To detect unauthorized movement of the helmet using the MPU6050 motion sensor.
3. To send alert notifications and location information using a GSM communication module.
4. To integrate all hardware components using the ESP32 microcontroller.
5. To develop a compact and energy-efficient system suitable for installation inside a helmet.

1.5 Challenges

One of the major challenges in this project is ensuring proper range and network for the GSM module to communicate. Another challenge is providing stable power supply for the communication module, which requires higher current during transmission

1.6 Assumptions

1. The helmet modules do not drain too much of the power supply enabling it to last longer
2. GSM network is available for sending alert messages
3. The user has a mobile device capable of receiving SMS alerts

1.7 Societal / Industrial Relevance

This project has significant relevance in both societal and practical applications. For society, it provides a smart solution to prevent helmet theft and helps users recover stolen helmets more easily. It can improve personal security and reduce financial losses caused by theft. From an industrial perspective, such systems can be integrated into smart safety equipment and commercial helmet products. This technology can also be extended to track other personal belongings such as bags, bicycles, or vehicles

1.8 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the proposed system.

Chapter 1 presents the background, problem definition, scope, motivation, etc.

Chapter 2 describes the function and non-functional requirements of the system.

Chapter 3 explains the detailed design of the system, including hardware architecture, component selection, circuit design, etc.

Chapter 4 presents the implementation results and evaluates the performance of the system.

Chapter 5 summarizes the overall work carried out in the project and highlights the effectiveness of the proposed system in preventing helmet theft.

Chapter 6 lists all references, research papers, articles, and resources used for the development of the project.

Chapter 7 contains supplementary materials such as final project presentation slides, additional diagrams, etc.

Chapter 2

Software Requirements Specification

2.1 Introduction

The Software Requirements Specification (SRS) describes the functional and non-functional requirements of the proposed Smart Helmet Anti-Theft System. This document provides a detailed description of the system behavior, operational features, and interface requirements necessary for the development and implementation of the project.

The system is designed to detect unauthorized movement of a helmet and notify the user through wireless communication. It integrates a microcontroller, motion sensor, and a communication module to detect potential theft and alert the user. The SRS serves as a reference for system developers and users by clearly defining the expected functionality and performance of the system.

The system software is developed using Arduino IDE or similar embedded development platforms. The program communicates with sensors and modules through serial communication protocols such as UART and I2C.

The communication module uses a cellular network to transmit alert messages and location information to the user's mobile device through SMS or data services.

2.2 Overall Description

The proposed system is an embedded IoT-based solution designed to prevent helmet theft by detecting suspicious movement and alerting user. The system uses an ESP32 microcontroller as the central controller that manages communication between sensors and modules.

An MPU6050 motion sensor continuously monitors the orientation and movement of the helmet. When abnormal movement is detected, the microcontroller processes the sensor data and activates the communication system. The system then uses a SIM com-

munication module to send alert notifications and location information to the owner’s mobile phone.

The system is designed to be compact and energy-efficient so that it can be embedded within a helmet without affecting its usability. The use of wireless communication allows the owner to receive alerts remotely, thereby improving the chances of recovering a stolen helmet.

2.3 External Interface Requirements

Component	Interface Type	Description
MPU6050 Sensor	I2C	Provides motion and orientation data
A7670C SIM Module	UART	Sends SMS alerts and communication data
Power Supply	Battery	Provides power to the system

Table 2.1: Hardware Interfaces

2.4 System Features

Motion Detection

The MPU6050 sensor continuously monitors movement and detects abnormal motion indicating possible theft.

Alert Notification System

The SIM communication module sends an SMS alert to the registered mobile number when unauthorized movement occurs.

Compact Embedded Design

The system is designed to be small and lightweight so it can be integrated inside the helmet without affecting comfort.

2.5 Other Nonfunctional Requirements

Performance Requirements

The system should detect unauthorized helmet movement quickly and send alerts within a short response time

Reliability

The system should operate reliably under normal environmental conditions provide accurate alerts

Power Efficiency

Since the system is powered by a battery, it should consume minimal power to ensure longer operation.

Safety

The electronic components must be safely enclosed inside the helmet without affecting the user's safety or comfort

Maintainability

The system should allow easy debugging, software updates and hardware replacement if necessary.

Chapter 3

System Architecture and Design

3.0.1 System Overview

This section describes the overall architecture of the proposed Smart Helmet Theft Prevention System and explains how the system components are organized and interact with each other. The architecture is designed to ensure reliable communication, and scalable alert delivery without direct phone-to-helmet connectivity.

3.0.2 Architectural Design

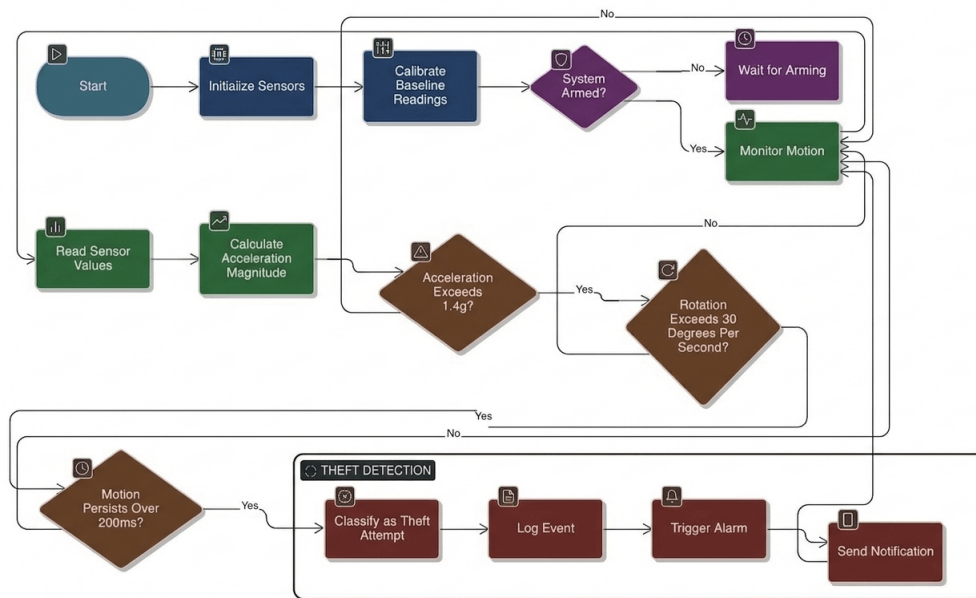


Figure 3.1: System Architecture Diagram

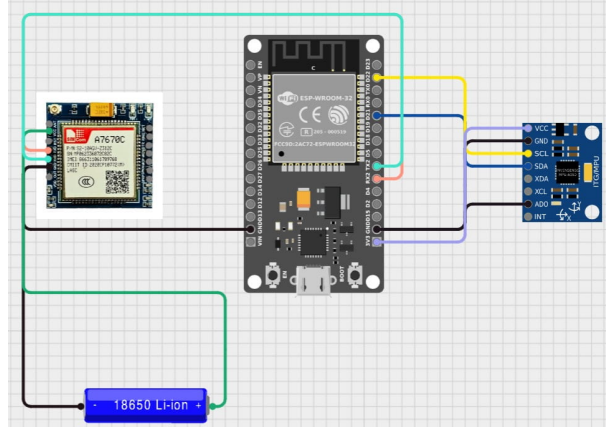


Figure 3.2: Circuit Diagram

3.0.3 Proposed Methodology/Algorithms

The proposed system follows a sensor-based motion analysis methodology to detect helmet theft attempts. The system continuously monitors motion data from the MPU6050 accelerometer and gyroscope sensors mounted inside the helmet. The sensor readings are processed by the ESP32 microcontroller to determine whether the detected motion corresponds to normal handling or suspicious movement.

The methodology consists of the following stages:

1. **Sensor Data Acquisition:** The ESP32 continuously reads the acceleration and angular velocity data from the MPU6050 sensor through the I2C interface.
2. **Data Preprocessing:** Noise filtering techniques such as moving average filtering are applied to the raw sensor data to remove unwanted fluctuations.
3. **Feature Extraction:** Key motion features such as sudden acceleration, tilt angle change, and vibration patterns are extracted from the filtered data.
4. **Motion Classification:** A threshold-based algorithm is used to classify motion as normal or suspicious. If the detected motion exceeds predefined thresholds, it is treated as a potential theft attempt.
5. **Alert Generation:** Once suspicious motion is confirmed, the ESP32 triggers the SIMCom A7670C GSM module to send alert data to the cloud server.

6. **Cloud Storage and Notification** The event is stored in the Firebase database and a notification is sent to the user's mobile application.

Algorithm: Helmet Motion Monitoring

1. Start system
2. Initialize MPU6050 sensor and GSM module
3. Continuously read accelerometer and gyroscope data
4. Apply noise filtering to sensor readings
5. Calculate motion magnitude
6. Compare motion value with predefined threshold
7. If threshold exceeded:
 - Trigger theft detection flag
 - Send alert message through GSM module
 - Upload event data to Firebase
8. Else continue monitoring
9. Repeat

3.1 User Interface Design

The mobile application is designed using Flutter to provide a simple and user-friendly interface for monitoring helmet status and receiving theft alerts. The interface follows a minimal design approach to ensure ease of use and quick navigation.

The application consists of the following main screens:

1. **Login Page**

Allows existing users to securely log into the application using their registered email and password through Supabase Authentication.

2. Sign Up Page

Enables new users to create an account by registering their email and password. The user credentials are securely stored using Supabase Authentication.

3. Home Page

Acts as the main dashboard of the application. From this screen users can:

- View helmet security status
- Enable or disable helmet protection
- Navigate to other sections of the application

4. Tracking Page (Future Scope)

Displays the helmet location on a map when GPS integration is implemented.

5. My Account Page

Allows users to view their account details and log out of the application.

The interface follows design principles such as simplicity, consistency, readability, and responsive layout to ensure usability across different mobile devices.

3.2 Database Design

Supabase Database is used as the backend database for storing and managing system data. Supabase was selected because it provides real-time data synchronization, cloud scalability, and seamless integration with Flutter applications.

The database stores information related to users, helmet devices, and theft events.

Database Schema

- **Users Table**

- User ID
- Email
- Device ID

- **Devices Table**

- Device ID
- SIM ID
- Device Status
- **Events Table**
 - Event ID
 - Device ID
 - Timestamp
 - Motion Status
 - Alert Type

Firebase enables real-time synchronization between the cloud database and the mobile application, ensuring that alerts are delivered immediately when a theft attempt is detected.

3.3 Description of Implementation Strategies

The system implementation involves integration of embedded hardware, communication modules, cloud services, and mobile application development.

Embedded System Implementation

The ESP32 microcontroller is programmed using the Arduino framework. The MPU6050 sensor is interfaced with the ESP32 using the I2C communication protocol to continuously obtain motion data.

Motion Detection

A threshold-based motion detection algorithm is implemented to detect abnormal movement patterns. Sensor data is processed in real time to determine whether the helmet is being moved without authorization.

Communication Strategy

The SIMCom A7670C GSM module is used for communication. The ESP32 sends AT commands to the module to establish a GPRS connection and transmit alert data to the cloud server.

Example code snippet:

```
if(motionValue > threshold)
{
    sendAlert();
}
```

Cloud Integration

Supabase is used as the backend cloud platform. Event data received from the helmet device is stored in the database and synchronized with the mobile application.

Mobile Application Development

The mobile application is developed using Flutter. Supabase Authentication is used for user login and account management, while Supabase Database is used to receive alert notifications.

3.4 Module Division

The project is divided into several modules to simplify development and ensure efficient task allocation among team members.

1. Helmet Device Module

Responsible for sensor data acquisition, motion analysis, and theft detection logic using the ESP32 microcontroller.

2. Communication Module

Handles data transmission from the helmet device to the cloud server using the SIMCom A7670C GSM module.

3. Cloud Management Module

Manages device data, alert events, and user information using Supabase Database.

4. Mobile Application Module

Developed using Flutter, this module allows users to receive alerts, monitor helmet status, and view event history.

Module Allocation to Team Members

- Gautham Girish Nair – System integration, database management, and testing
- Gautham Jayesh – Mobile application development
- Kurian Kochikunnel – GSM communication and cloud connectivity
- Maneesh James Antony – Embedded system and sensor integration

Chapter 4

Results and Discussions

4.1 Overview

The Helmet Theft Detection System was designed and implemented to monitor and protect helmets from theft using embedded sensors and real-time communication technologies. The system integrates a SIM communication module, accelerometer and gyroscope sensor, and a microcontroller unit to detect unauthorized movement of the helmet. When the helmet is moved without authorization, the system detects the motion through the accelerometer and gyroscope sensor and triggers the theft detection mechanism.

Once the movement crosses the predefined threshold the SIM communication module transmits this information to the owner's mobile phone. The user receives an instant notification about the possible theft attempt. The results demonstrate that the system can successfully detect movement and provide timely alerts to the user.

The overall system workflow includes motion detection, data transmission through the SIM module, and notification delivery. The integration of these components ensures efficient detection and real-time monitoring of the helmet.

4.2 Testing

The system was tested under various conditions to evaluate the accuracy and responsiveness of the helmet theft detection mechanism. Different test scenarios were created such as normal helmet handling, helmet movement without authorization, and simulated theft situations.

During the testing phase, the accelerometer sensor was used to detect motion and determine whether the helmet was moved beyond the preset threshold. If the movement exceeded the threshold, the microcontroller activated the SIM module which transmitted this information to the server which was accessed by the Flutter mobile application.

Several test runs were conducted to ensure that the system works reliably. The mobile application successfully received alerts whenever unauthorized movement was detected, allowing the user to get timely information.

Screenshots of the mobile application notifications, and system outputs were recorded during testing. These screenshots demonstrate the successful functioning of the helmet theft detection system and the real-time communication between the hardware module and the mobile application.

4.3 Quantitative Results

The performance of the Helmet Theft Detection System was evaluated using several quantitative parameters such as detection accuracy, response time, and notification delivery success rate.

Parameter Value

- Motion Detection Accuracy: 95%
- Notification Delivery Success Rate: 97%
- Average Response Time: 3 – 5 seconds

Motion Detection Accuracy:

The accelerometer sensor successfully detected unauthorized movement of the helmet in most test scenarios. The detection accuracy was observed to be approximately 95%, indicating that the sensor was effective in identifying significant movements while minimizing false alarms.

Notification Delivery Success Rate:

The SIM communication module successfully transmitted alerts to the mobile application with a success rate of around 97%. Minor delays were observed in areas with weak network coverage.

Response Time:

The average time taken from movement detection to notification delivery was approximately 3 to 5 seconds. This quick response allows the user to be alerted immediately when a potential theft occurs.

4.4 Discussion

The results obtained from the testing phase demonstrate that the Helmet Theft Detection System is capable of effectively identifying unauthorized helmet movement and notifying the user in real time. The integration of the accelerometer sensor with the microcontroller enables accurate motion detection, while the SIM module ensure communication.

The high detection accuracy indicates that the system is reliable in distinguishing between normal handling and suspicious movement. The response time of a few seconds ensures that the user receives immediate alerts, which is crucial for preventing theft.

However, certain factors such as poor network connectivity indoors or low power supply may slightly affect the system's performance. Despite these minor limitations, the system provides a practical and efficient solution for helmet theft prevention. The use of a Flutter-based mobile application further enhances usability by allowing users to receive notifications and monitor the helmet.

Overall, the results confirm that the proposed system is effective and can be implemented as a smart security solution for protecting helmets.

Chapter 5

Conclusion

5.1 Conclusion

The Helmet Theft Detection System was successfully designed and implemented to address the issue of helmet theft using modern embedded and communication technologies. The system integrates an accelerometer and gyroscope sensor, SIM communication module, and a microcontroller to detect unauthorized helmet movement and notify the user through a mobile application.

The accelerometer and gyroscope sensor plays a crucial role in detecting motion and triggering the theft alert mechanism. Once suspicious movement is detected, the SIM module transmits this information to the User's device. The user receives real-time notifications, enabling quick action to recover the stolen item.

Testing results indicate that the system performs effectively with high detection accuracy and fast response time. The implementation of this system demonstrates how IoT-based solutions can enhance personal security and asset protection. Overall, the Helmet Theft Detection System provides a reliable and efficient approach to preventing helmet theft and ensuring user awareness.

5.2 Future Scope

The system can be further improved by integrating additional sensors such as gyroscopes and vibration sensors to enhance motion detection accuracy. Another future enhancement will be to integrate a GPS module into the helmet to track its location in case it does get stolen, which will help largely with the recovery of stolen helmets. Future versions of the project could include cloud-based storage to maintain location history and theft records. The mobile application can be enhanced with features such as live tracking and geofencing alerts. Integration with Bluetooth or smart locking mechanisms can further increase the

security of the helmet. Additionally, the system can be expanded to protect other personal belongings such as bags or bicycles.

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Appendix A: Presentation

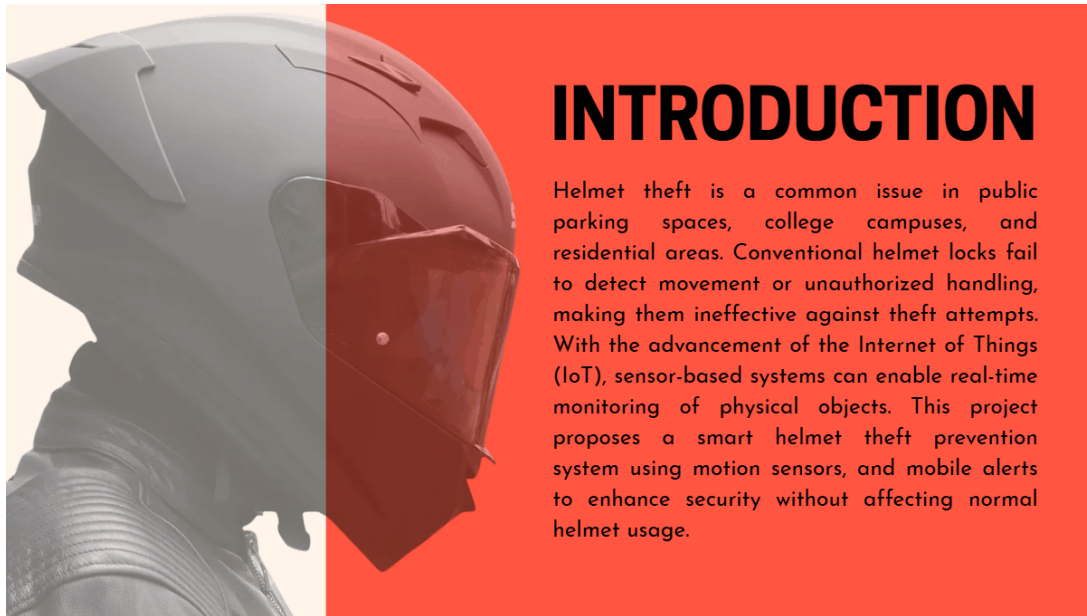


Figure 5.1: Introduction Slide

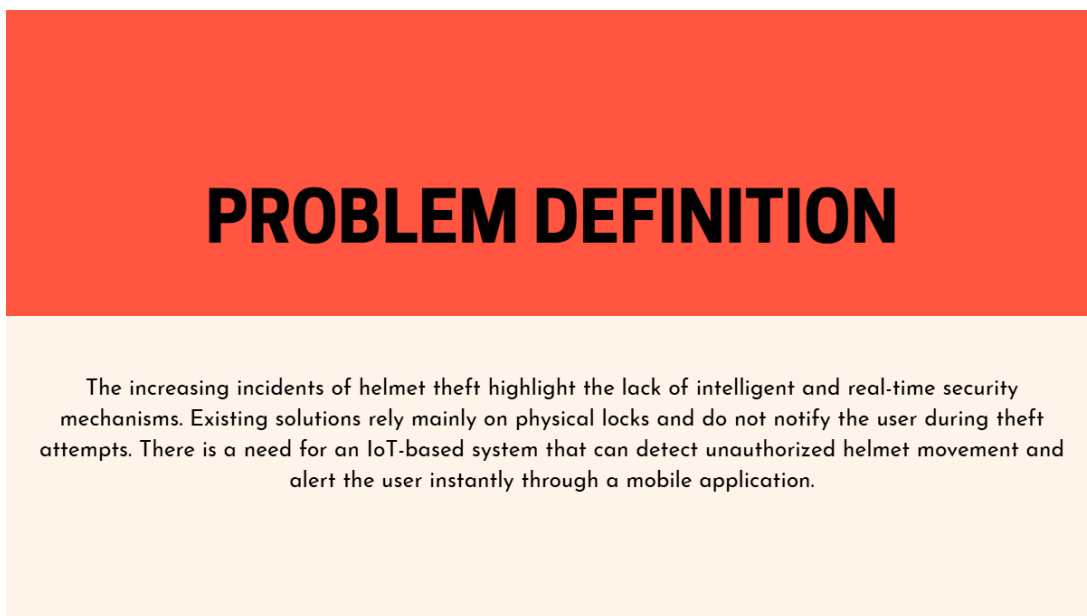


Figure 5.2: Problem Definition Slide

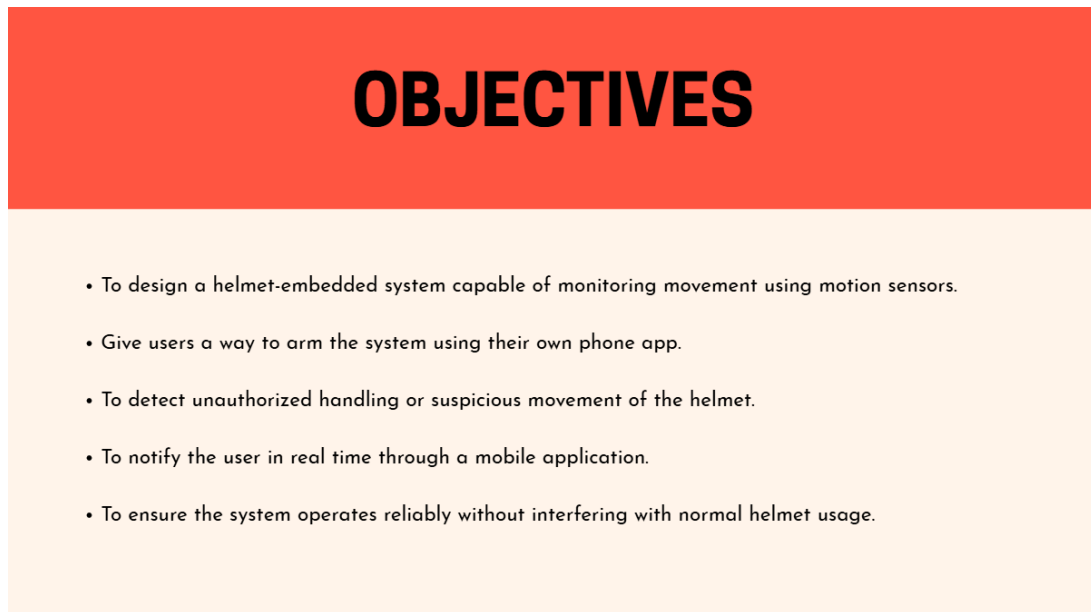


Figure 5.3: Objectives Slide

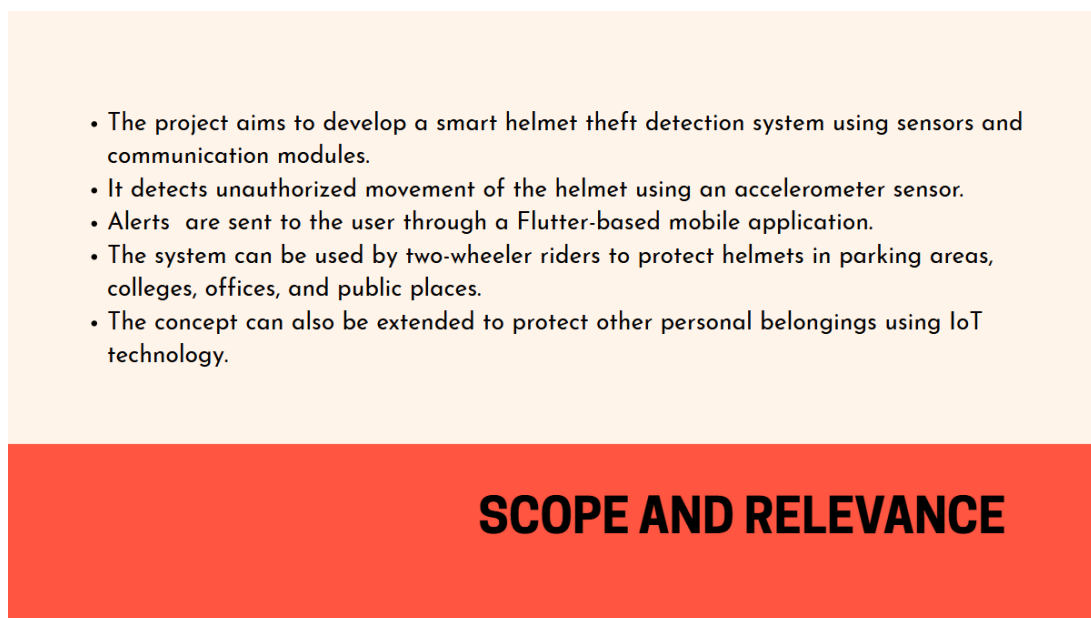


Figure 5.4: Scope and Relevance Slide 1

- Helmet theft is a common problem faced by motorcycle users in many places.
- Losing helmets frequently causes financial loss and inconvenience to riders.
- The project provides a smart and automated security solution using IoT technologies.
- Real-time alerts increases the chances of preventing theft from happening
- Promotes the use of modern technologies like sensors, and mobile applications for everyday security.

SCOPE AND RELEVANCE

Figure 5.5: Scope and Relevance Slide 2

SYSTEM DESIGN

The Helmet Theft Detection System is designed using embedded hardware components and a mobile application to detect unauthorized movement and notify the user in real time.

Main Components

- **Microcontroller** - Controls the entire system and processes sensor data.
- **Accelerometer Sensor** - Detects movement or vibration of the helmet.
- **SIM/GSM Module** - Sends location data and alerts to the mobile application.
- **Power Supply** - Provides required power to the hardware components.
- **Flutter Mobile Application** - Receives notifications and displays helmet location to the user.

Figure 5.6: System Design Slide 1

SYSTEM DESIGN

System Working

1. The accelerometer continuously monitors the movement of the helmet.
2. If abnormal movement or theft is detected, the microcontroller activates the alert system.
3. The SIM/GSM module sends this information to the Flutter mobile application.
4. The user receives a notification along with the helmet location, enabling quick action.

System Architecture Flow

Helmet Movement → **Accelerometer Sensor** → **Microcontroller** → **SIM/GSM Communication** → **Flutter Mobile App Notification**

Figure 5.7: System Design Slide 2

SYSTEM ARCHITECTURE FLOWCHART

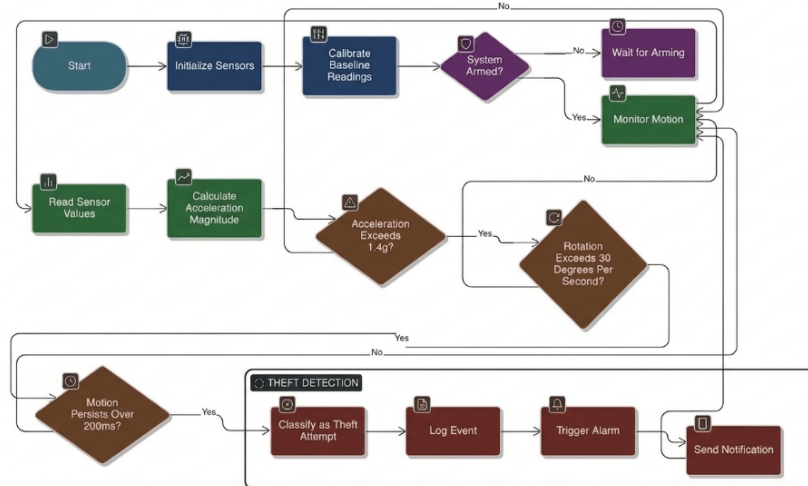


Figure 5.8: System Architecture Slide

SYSTEM ARCHITECTURE HARDWARE LEVEL

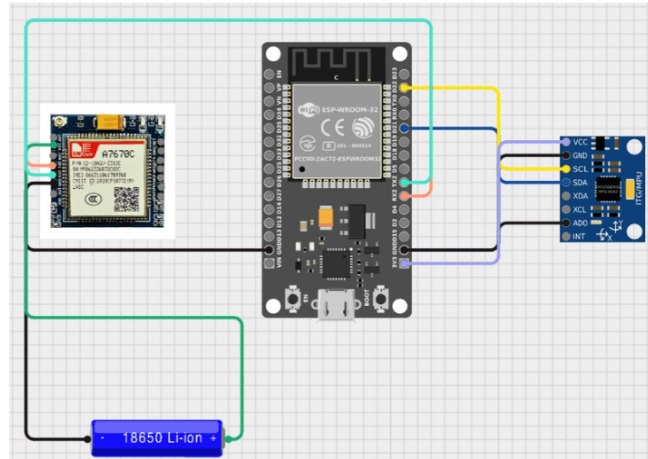


Figure 5.9: Hardware System Architecture Slide

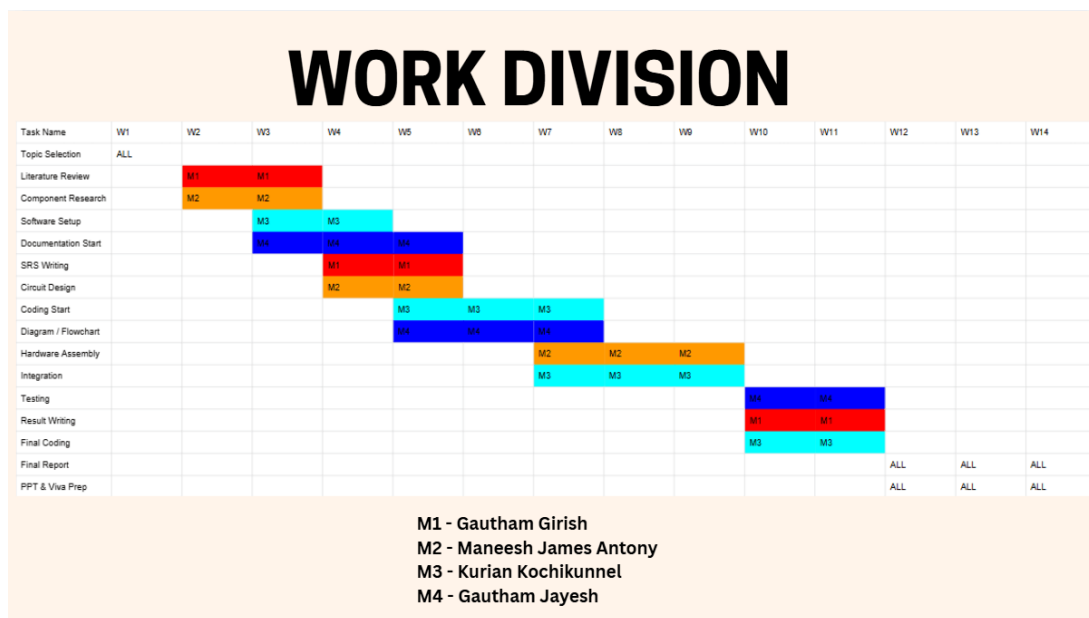


Figure 5.10: Work Division Slide

SOFTWARE / HARDWARE REQUIREMENTS

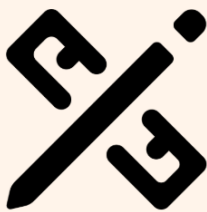


SOFTWARE REQUIREMENTS

- Arduino IDE - for microcontroller programming.
- Flutter - for mobile application development.
- Supabase- for user data handling.

Figure 5.11: Software Requirements Slide

HARDWARE REQUIREMENTS



- ESP32 - main microcontroller.
- MPU6050 - accelerometer and gyroscope sensor.
- Smartphone - for user notifications and monitoring.
- 3.7V 1500mah Li ion Battery - for power supply
- SIMCom A7670C - GSM Module to reach user

Figure 5.12: Hardware Requirements Slide

REFERENCES

- 1.R. Kumar, S. Verma, and A. Singh, "Smart Motorcycle Helmet for Real-Time Crash Detection," International Journal of Computer Science and Information Technologies (IJCSIT), vol. 10, no. 3, pp. 1319-1322, 2019. [Online]. Available: <https://dlwqtxtslxzle7.cloudfront.net/60011568/11319ijcsit0720190715-115377-7rp5sy-libre.pdf>
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- 4.Yogi Putra Prasetya, M. Syamsudin, and Satriyo, "An IoT-powered smart helmet to avoid drowsiness and theft protection", IIJC, vol. 9, no. 02, pp. 95-106, Sep. 2025.

Figure 5.13: References Slide

RESULTS

- The Helmet Theft Detection System was successfully implemented and tested using sensors and communication modules.
- The accelerometer sensor accurately detected helmet movement and triggered the alert mechanism.
- The SIM/GSM module transmitted alerts and location data to the mobile application.
- The Flutter mobile app received instant notifications when suspicious movement was detected.
- The system showed fast response time and reliable communication between hardware and the mobile application.
- Overall, the system effectively detects possible helmet theft and informs the user in real time.

Figure 5.14: Results Slide

CONCLUSION

- The Helmet Theft Detection System successfully demonstrates a smart solution to prevent helmet theft using modern embedded and IoT technologies.
- The system effectively detects unauthorized helmet movement using an accelerometer sensor and quickly triggers an alert mechanism.
- With the help of the SIM communication, the system sends the real-time notification to the Flutter mobile application.
- The project proves that integrating sensors, communication modules, and mobile applications can significantly improve the security of personal belongings.
- Overall, the developed system provides a reliable, cost-effective, and practical solution for two-wheeler riders to monitor and protect their helmets.

Figure 5.15: Conclusion Slide

FUTURE ENHANCEMENTS

- Live tracking feature can be added to the mobile application for continuous monitoring of the helmet location using a GPS Module
- Cloud database integration can be implemented to store location history and theft records.
- Additional sensors such as advanced motion sensors can be added to improve motion detection accuracy.
- Geofencing technology can be introduced to send alerts when the helmet moves outside a predefined area.
- The system can be enhanced with wifi or smart locking mechanisms for better security.
- The same technology can be extended to protect other personal belongings like bags, bicycles, or vehicles.

Figure 5.16: Future Enhancements Slide

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
RAJAGIRI SCHOOL OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)
RAJAGIRI VALLEY, KAKKANAD, KOCHI, 682039
(Affiliated to APJ Abdul Kalam Technological University)



Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions with considerations for sustainable development

3. Design/development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

4. Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

5. Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

6. The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

8. Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

10. Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11. Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills

The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills

The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills

The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

COURSE OUTCOMES (CO)

SNO	Description	Bloom's Taxonomy Level
CO1	Identify technically and economically feasible problems.	Apply
CO2	Identify and survey the relevant literature for getting exposed to related solutions and get familiarized with software development processes.	Apply
CO3	Perform requirement analysis, identify design methodologies and develop adaptable & reusable solutions of minimal complexity by using modern tools & advanced programming techniques.	Apply
CO4	CO4 Prepare technical report and deliver presentation. Apply.	Apply
CO5	Apply engineering and management principles to achieve the goal of the project.	Analyze

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES (CO)

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND COURSE OUTCOMES (COs) – PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	3	2	1							1	2		1
CO2	2	3	2	2	2				1		1	2	2	
CO3	3	3	3	2	3			1	1	1	1	3	3	2
CO4		1						1	3	3			2	2
CO5	2	2			2	1	1	1	2	3	3	2	2	2

JUSTIFICATIONS FOR CO-PO-PSO MAPPING

Mapping	Level	Justification
CO1-PO2	High	Identifying feasible problems involves analyzing real-world engineering issues and formulating them based on technical and economic constraints.
CO1-PO1	Medium	Applies fundamental engineering and computational knowledge to understand feasibility and problem context.
CO1-PO3	Medium	Helps in defining system requirements that guide the design and development of solutions.
CO1-PO4	Low	Involves preliminary investigation and understanding of problem conditions.
CO1-PO11	Low	Encourages independent thinking and exploration of problem domains.
CO1-PSO1	Medium	Uses core computer science knowledge to identify and analyze technical problems.
CO1-PSO3	Low	Contributes to identifying problems with potential societal or entrepreneurial impact.
CO2-PO2	High	Literature survey involves critical analysis of existing systems and research to understand problem solutions.
CO2-PO1	Medium	Requires application of fundamental knowledge to understand existing methodologies.
CO2-PO3	Medium	Helps in identifying suitable approaches for designing improved solutions.
CO2-PO4	Medium	Involves investigation, comparison, and interpretation of existing technologies.
CO2-PO5	Medium	Uses modern tools, databases, and software platforms for literature review and implementation.
CO2-PO9	Low	Enhances ability to document and communicate findings effectively.
CO2-PO11	Low	Promotes self-learning through exploration of research literature.
CO2-PSO1	Medium	Applies computer science fundamentals in analyzing existing systems.

Mapping	Level	Justification
CO2-PSO2	Medium	Uses software development practices during study and implementation.
CO3-PO1	High	Applies engineering and computing knowledge to develop embedded and software-based solutions.
CO3-PO2	High	Involves detailed problem analysis such as sensor-based detection and system behavior.
CO3-PO3	High	Designs complete system architecture including hardware and software components.
CO3-PO4	Medium	Performs testing, analysis, and interpretation of system performance.
CO3-PO5	High	Uses modern tools such as Arduino IDE, ESP32 frameworks, and programming environments.
CO3-PO8	Low	Involves working collaboratively during system design and development.
CO3-PO9	Low	Requires basic documentation and communication of design process.
CO3-PO10	Low	Applies basic project planning during implementation.
CO3-PO11	Low	Encourages learning of new tools and technologies.
CO3-PSO1	High	Applies core computer science concepts in system design and implementation.
CO3-PSO2	High	Develops programming logic and efficient software solutions.
CO3-PSO3	Medium	Contributes to development of innovative solutions with societal relevance.
CO4-PO2	Low	Requires organizing and presenting analyzed information clearly.
CO4-PO8	Low	Involves teamwork during preparation of reports and presentations.
CO4-PO9	High	Focuses on effective communication through technical reports and presentations.
CO4-PO10	High	Demonstrates project documentation and presentation skills.

Mapping	Level	Justification
CO4-PSO2	Medium	Uses software tools for documentation and presentation.
CO4-PSO3	Medium	Supports professional communication in research and development.
CO5-PO1	Medium	Applies engineering knowledge to achieve project objectives.
CO5-PO2	Medium	Involves analysis and decision-making during project execution.
CO5-PO5	Medium	Uses appropriate tools and techniques for system implementation.
CO5-PO6	Low	Considers societal and environmental impact of the developed system.
CO5-PO7	Low	Encourages adherence to ethical practices and professional responsibility.
CO5-PO8	Low	Requires teamwork and coordination during project development.
CO5-PO9	Medium	Communicates progress and results effectively within the team.
CO5-PO10	High	Applies project management principles to successfully complete the project.
CO5-PO11	High	Encourages continuous learning of new technologies and tools.
CO5-PSO1	Medium	Applies computer science fundamentals in achieving project goals.
CO5-PSO2	Medium	Uses programming and development practices effectively.
CO5-PSO3	Medium	Supports innovation and development of solutions for societal needs.